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SUMMARY

AI/ML Engineer with a BS in Computer Engineering (Gold Medalist, CGPA 3.94/4.00) and four internships building LLM systems, agentic workflows, and RAG pipelines in Python and Node.js. Currently building LLM fine-tuning, quantization, and serving pipelines (QLoRA, AWQ, vLLM, FastAPI) with benchmark-validated results. Experience spans real-time conversational AI agents, multi-agent orchestration (LangChain/LangGraph), backend API engineering, and applied computer vision.

PROJECTS

FT-Bench: End-to-End LLM Fine-Tuning & Quantization Benchmark

Llama-3.2-3B • QLoRA • AWQ • vLLM

- Fine-tuned Llama-3.2-3B-Instruct with QLoRA (rank-16 adapters, 4-bit NF4 base) on 11,481 MASSIVE NLU samples, raising exact match from 2.62% to 71.69% and Slot F1 from 16.41% to 85.34% on 2,974 held-out samples (bootstrap CIs, n=1,000).
- Applied AWQ INT4 quantization (group size 128, llm-compressor), retaining 72.19% exact match (+0.5 pp vs. FP16) while cutting VRAM from 6.2 GB to 3.8 GB; modeled self-hosting at ~93% lower cost than GPT-4o API at 100K queries/day.
- Built an evaluation and serving pipeline with FastAPI, vLLM (PagedAttention), 4-quadrant error analysis, 34-unit tests, GitHub Actions CI/CD, and a live Streamlit dashboard; published 3 Hugging Face checkpoints (adapter, FP16, AWQ INT4).

AI-Driven Multi-Agent System for Disaster Response

1st Place, Dept. Open House Exhibition

- Built a multi-agent simulation using Deep Reinforcement Learning (Stable-Baselines3) on OpenStreetMap-derived road networks, modeling emergency-response coordination across 50+ simulated nodes.
- Measured a 15–20% reduction in simulated response time versus an uncoordinated baseline; built a real-time Streamlit and Plotly dashboard to visualize agent behavior and performance metrics.

AI Travel Planner Agent

LangChain • RAG • Python

- Built an autonomous travel-planning agent with LangChain and Python to process multi-constraint, multi-turn itineraries, tracking and resolving 10+ sequential user constraints.
- Implemented RAG with a vector database to ground agent responses in real-time travel data.

WORK EXPERIENCE

AI Intern / Artificizen, Onsite

Jan 2026 – Jul 2026

- Built real-time conversational AI calling agents with Retell, Dograh (containerized and deployed with Docker), and LiveKit, orchestrating Gemini and Grok LLMs through LangChain and LangGraph for multi-step agentic workflows.
- Implemented FastAPI backend services on MySQL/Neon, integrating FAISS and ChromaDB vector stores for retrieval in live calling workflows.
- Tested and validated agent workflows across client projects; authored technical documentation of system architecture for stakeholders.

Full-Stack Developer Intern / Pinnacloid, Onsite

Aug 2025 – Oct 2025

- Built full-stack features in Node.js and React.js to integrate LLMs into production environments.
- Optimized 12+ REST API endpoints and data retrieval logic for real-time RAG workflows, reducing measured request latency by 25%.
- Resolved 30+ production bugs and led Node.js backend orchestration for cross-team feature integration.

Machine Learning Intern / Arch Technologies, Remote

Jun 2025 – Jul 2025

- Adapted LLaMA 3.2 (3B) to proprietary healthcare data using prompt engineering and medical chain-of-thought datasets.
- Built a brain tumor detection and segmentation pipeline with YOLO11 and SAM2, with model evaluation integrated into the workflow.
- Built agentic workflows for medical data processing, including voice-cloning pipelines and vector-database-backed retrieval.

React Developer / YoungDevIntern Company, Remote

Sep 2024 – Oct 2024

- Built 3+ React.js, Redux, and Tailwind CSS interfaces for interacting with datasets and AI workflows in Agile/Scrum teams.

EDUCATION

BS in Computer Engineering / The University of Lahore, Lahore

Sep 2022 – Jul 2026

- Gold Medalist; ranked #1 in the Computer Engineering batch. CGPA: 3.94/4.00.

SKILLS

LLM & Generative AI: LLM Fine-Tuning (QLoRA/LoRA, PEFT), AWQ Quantization, vLLM Serving, Hugging Face Transformers, RAG, Agentic Orchestration, LangChain, LangGraph, Prompt Engineering

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, NLP, Computer Vision (YOLO11, SAM2), Reinforcement Learning (Stable-Baselines3), Model Evaluation

Backend & MLOps: FastAPI, REST APIs, Docker, GitHub Actions (CI/CD), Vector Databases (FAISS, ChromaDB), MySQL, Neon (Postgres)

Voice & Real-Time AI: Retell, Dograh, LiveKit

Languages & Tools: Python, JavaScript, C++, SQL, Node.js, React.js, Streamlit, Git

CERTIFICATIONS & LANGUAGES

Certifications: Coursera: Deep Learning Specialization; Machine Learning Specialization; Advanced React | freeCodeCamp: Data Analysis & Scientific Computing with Python; JavaScript Algorithms & Data Structures

Languages: English (Proficient) • Tamil (Native) • Sinhala (Intermediate) • Urdu (Intermediate)